

Do Birds of Father Flock Together? Examining Homophily Dynamics at a Religious University

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Abstract

Traditional homophily indices often conflate individual preferences with baseline structural opportunities, creating systematic biases when comparing minority and majority groups. We address this by examining religious homophily and friendship choice using longitudinal network data from the NetHealth Cohort Study ($N = 722$ students). Using a dyadic framework with a mathematical opportunity offset, we adjust for baseline group sizes and physical and social foci, such as residential proximity and demographic matching. We find that all religious groups exhibit persistent inbreeding homophily. However, our opportunity-adjusted models reveal that minority religious affiliation (e.g., Protestant and other religions) has a stronger positive effect on active homophily compared to the Catholic majority, supporting Minority Distinctiveness Theory. Conversely, secular “No Religion” clustering is substantially explained by physical and demographic foci, highlighting a key theoretical distinction between active and structurally induced homophily in highly homogeneous social environments.

1 Introduction

1.1 Background

How does the numeric composition of a local social environment shape the formation of friendship ties across religious groups? While the robust tendency for individuals to associate with similar others—homophily—is one of social science’s most well-established empirical generalizations (McPherson et al., 2001), explaining *why* different groups exhibit varying degrees of homophily within a single population remains an open area of research. Although selectively forming ties with similar others can be an individual choice, it is also influenced by structural factors such as propinquity, opportunity structure, and social foci (Rivera et al., 2010; Feld and Grofman, 2009). In this regard, network analysts distinguish between “choice homophily” resulting from individual preferences and “induced homophily” resulting from structural processes such as uneven group sizes or physical proximity (Kossinets and Watts, 2009; McPherson et al., 2001; Marsden, 1988; Hofstra et al., 2017).

A subset of structural theories of homophily focuses on how the uneven distribution of group sizes creates motivational (dis)incentives and a biased opportunity structure landscape for forming same-group and out-group ties. In this regard, there are two competing explanations for how relative group size intersects with homophily (and heterophily) generating processes. Under the *Minority Distinctiveness Hypothesis* (MDH), a numeric minority status increases the cognitive salience of that group identity in the context of outgroup numerical dominance, inducing active, deliberate sorting among minority members who band together to preserve their subcultural boundaries (Mehra et al., 1998; Leonard et al., 2008; Marsden, 1988; Sepulvado et al., 2015). Conversely, following Blau’s

(1977a) classic treatment, what we might refer to as *Majority Exclusiveness Hypothesis* (MEH) suggests that a dominant majority possesses the structural and social leverage to guard group boundaries, which means that they should exhibit stronger tendencies towards homophily doubly aided by choice (majority members are attractive partners in a context in which they are dominant) and “induced” (there are more members of the majority to choose from) considerations. As Blau observed, “the larger of two groups discriminates more than the smaller against associating with members of the other group,” resulting in higher active homophily among the majority (Blau, 1977a, p. 81).

1.2 Study Setting

An ideal setting in which to examine the MDH and MEH predictions would be a highly specific local context in which a single group holds overwhelming numerical dominance. The University of Notre Dame (ND) presents such a unique social laboratory for the present research. In the *NetHealth Study*’s (Purta et al., 2016) longitudinal cohort, Catholic students represent approximately 73% of the student body, while Protestant (9.9%), non-religious (12.4%), and other religious students (4.6%) constitute small, distinct minorities. However, this setting introduces a profound institutional focus (Feld, 1981; Feld and Grofman, 2009): religion is not merely a demographic attribute; it defines the university’s very identity and physical infrastructure. Crucially, Catholics attending a Catholic university might be more motivated to befriend other Catholics because the environment “summons” their religious identity, increasing its salience (Tavory, 2016). Furthermore, ND features built-in residential and social foci—such as single-sex dormitories, mandatory dorm assignments, and Catholic-oriented campus activities—that could organically channel Catholics into homophilous friendships without requiring active individual preference for Catholic friends (although such preferences undoubtedly exist in this context).

To evaluate the predictions of the MDH (higher homophily rates among minorities) versus the MEH (higher homophily rates among the majority), it is important to move beyond the limitations of raw homophily rates and aggregate indices. Traditional aggregate measures, such as Krackhardt’s E-I (External-Internal) index (Krackhardt and Stern, 1988), suffer from a well-known mathematical artifact: in a bounded network, a numeric minority is structurally forced to have higher outgroup contact, regardless of their personal preferences, a phenomenon also emphasized by Blau (1977b), thus producing a form of “induced heterophily.” Consequently, aggregate comparisons are biased (Everett and Borgatti, 2012). Here, we address that issue by transitioning to a dyadic (tie-level) framework with a mathematical opportunity offset. This approach allows us to adjust for baseline group sizes in the data, as well as other important drivers of homophily, such as physical propinquity and social foci, as well as other forms of homophily beyond the focal one in our study.

To this end, this paper leverages 8 waves of longitudinal network data from the *NetHealth Study* to examine the trajectory of religious homophily. We test whether the observed homophily of Protestants and other religious minorities is merely a byproduct of residential sorting and alternative tie-level demographic similarities, or whether it reflects a robust, active sorting process driven by identity salience amid Catholic numerical dominance, as predicted by the MDH.

2 Methods and Data

2.1 Data Source and Sample Design

We evaluate the dynamics of religious homophily and friendship choice using longitudinal social network and survey data from the *NetHealth Study* at the University of Notre Dame, a multi-

year project funded by the National Institutes of Health. NetHealth tracked a single cohort of undergraduate students from their entry as freshmen in Autumn 2015 through their graduation in Spring 2019. Initial recruitment enlisted around 700 students (yielding a total sample of $N = 722$ participants who took at least one survey), tracking their social networks, behaviors, and attributes over 8 waves of data collection fielded approximately every 3–4 months.

In each of the eight waves, respondents completed a name-generator survey with the prompt: “We would like to know who you consider to be in your social network. In the spaces below please list up to 20 people with whom you spend time communicating or interacting.” Beginning with Wave 3, respondents were presented with their previously nominated alters and allowed to add up to 5 of them, yielding a maximum of 25 alters. These name generators, coupled with name interpreters detailing the characteristics of each alter and the nature of the relationship, yielded a unique multiwave, long-term ego-network dataset that constitutes our primary data source. For the purposes of evaluating within-cohort homophily, we restrict alter nominations to those designated as fellow on-campus students.

2.2 Variables

2.2.1 Ego-Level (Participant) Measures

Ego-level attributes are drawn from the NetHealth Basic Survey, which captured demographic and behavioral characteristics across the eight waves. Baseline traits were measured at Wave 1.

- **Ego Religion:** The primary independent variable is baseline religious affiliation. Respondents were categorized into four groups: Catholic (72.7%, the dominant majority), No Religion (12.3%), Protestant (9.8%), and Other Religion (4.6%). Missing responses account for < 1%.
- **Ego Gender:** Respondent gender, coded as Female (51.1%) or Male (47.9%).
- **Ego Race/Ethnicity:** Respondent race/ethnicity, categorized into White (65.0%), Latino/a (12.3%), Asian-American (8.7%), Foreign Student (7.1%), African-American (5.7%), and Other (< 1%).
- **Religious Salience (Frequency of Discussion):** To control for individual-level religious salience independent of homophily, we use a measure of how frequently the respondent discusses religion (captured in Wave 2). Responses are recorded on a 5-point Likert scale ranging from “Not at all” (4.6%), “Less than 1-2 times a month” (11.8%), “1-2 times a month” (31.9%), “1-2 times a week” (27.4%), to “Three times a week or more” (12.6%).

2.2.2 Alter-Level (Tie) Measures

Alter characteristics and relationship contexts are derived from the NetHealth Network Survey, where egos provided information on their nominated peers.

- **Alter Relationship Context:** The relation of the alter to the university (e.g., Student, Alum, Prospect). Overall, 48.6% of nominations were for fellow students. We subset our dyadic data to these alters ($N = 17,452$ student ties).
- **Alter Religion:** Egos reported the perceived religious affiliation of each nominated fellow student (Catholic: 67.8%; Protestant: 6.8%; No Religion: 7.2%; Other Religion: 2.2%; Missing: 15.9%). This is used to construct our primary outcome variable, a binary indicator of religious homophily (coded 1 if ego and alter share the same religious affiliation, and 0 otherwise).

- **Alter Gender and Race:** Egos reported each student alter’s sex (Female: 52.2%; Male: 47.8%) and race/ethnicity (e.g., White: 78.0%). These allow us to construct binary dyadic indicators for gender matching and racial/ethnic matching.
- **Physical and Social Foci (Roommates and Dorms):** To adjust for structurally induced clustering, egos indicated whether each fellow student was a roommate (13.6% yes) and whether they resided in the same dormitory (34.9% yes). At Notre Dame, the dormitory system constitutes a primary axis of residential and social life.
- **Friendship Closeness (Intimacy):** Egos were asked to report how close they felt to each nominated alter on an ordered scale. We construct an ordered numeric indicator for friendship intensity (coded 1 for “Distant”, 2 for “Less Than Close”, 3 for “Merely Close”, and 4 for “Especially Close”). This continuous indicator is added to our multivariate models to control for baseline differences in the homophily of strong versus weak ties.

2.3 Mathematical Opportunity Offset Formulation

To model the likelihood of a tie being homophilous (where both individuals share the same religion) while adjusting for baseline group proportions, we employ a dyadic logistic regression with a mathematical opportunity offset. In highly skewed settings like Notre Dame, where Catholics make up nearly three-quarters of the student body, a traditional regression approach struggles to separate active friendship preferences from simple numeric availability. A Catholic student is highly likely to befriend another Catholic simply because the vast majority of their available peers are Catholic, not necessarily because they are actively seeking out a same-religion friend. The opportunity offset corrects for this by mathematically subtracting the baseline probability of meeting someone of the same religion under conditions of random mixing.

Formally, we specify the model as:

$$\text{logit}(P(\text{Same Religion} = 1)) = \beta_0 + \beta_1 \text{EgoReligion} + \mathbf{X}\beta + \text{opportunity_offset} \quad (1)$$

In this equation, the opportunity_offset is defined as $\log\left(\frac{p_w}{1-p_w}\right)$, where p_w represents the overall proportion of the respondent’s religious group in the available pool of peers for that specific wave. By forcing this term into the model with a fixed coefficient of 1.0, we adjust the baseline odds of tie formation. Consequently, the remaining coefficients represent the *excess* tendency to form a homophilous tie above and beyond what the structural opportunity pool dictates.

The model also includes $\mathbf{X}$, a vector of dyadic controls representing alternative mechanisms that might induce homophily without active religious sorting. These include demographic similarities (same gender and same race) as well as physical and social foci (being roommates or living in the same dorm).

Crucially, this specification provides a direct test of our competing theoretical frameworks. A positive β_1 coefficient for a minority group indicates that the group exhibits significantly stronger religious homophily than the Catholic majority, even after accounting for their smaller population size and alternative sorting mechanisms. Such a result would directly support the Minority Distinctiveness Hypothesis, demonstrating that numeric minorities actively forge tighter same-group bonds in response to a dominant majority environment.

3 Results

3.1 Baseline Opportunity Structure Stability

Before analyzing active homophily, it is crucial to establish the stability of the baseline opportunity structure. If the numeric composition of the student body fluctuated wildly over the four years, any changes in tie formation could simply be an artifact of changing demographics rather than evolving social preferences. The wave-specific proportions of available student peers across Waves 3 to 8 are plotted in Figure 1. The extreme flatness of these lines visually confirms that the structural opportunity pool is highly stable, ensuring that the changes in homophily observed in subsequent models reflect genuine behavioral sorting rather than demographic drift.

3.2 Longitudinal Trajectories of Religious Homophily (Yule’s Q)

To address the group-size artifact, we first evaluate the longitudinal trajectory of religious mixing using group-level, baseline-adjusted Yule’s Q across all eight waves of the college cohort (see Figure 2). Unlike alternative aggregate metrics—such as raw in-group tie percentages, Coleman’s homophily index, or Krackhardt and Stern’s E-I (External-Internal) index—Yule’s Q is mathematically independent of relative group sizes. Because traditional measures like the E-I index are strictly bounded by the available opportunity pool, they inherently force numeric minorities to exhibit mathematically higher rates of heterophily simply due to their small baseline population, distorting cross-group comparisons. In contrast, Yule’s Q is a direct transformation of the odds ratio, rendering it margin-free. This ensures that the metric captures active sorting dynamics independent of the severe group size imbalances in our sample, allowing for rigorous and unbiased comparisons between the Catholic majority and various minority groups.

Yule’s Q is defined as:

$$Q = \frac{ad - bc}{ad + bc} \quad (2)$$

where a is the number of ingroup ties (e.g., Catholics befriending Catholics), b is the number of outgroup ties from group G (Catholics befriending non-Catholics), c is ties from the outgroup to group G (non-Catholics befriending Catholics), and d is ties strictly between outgroup members (non-Catholics befriending non-Catholics).

The resulting index ranges from -1 to $+1$. A value of 0 indicates perfect random mixing, where individuals choose friends without any regard to religion. A positive value ($Q > 0$) indicates “inbreeding homophily,” meaning individuals actively prefer same-group friends at higher rates than chance would dictate. At the extreme, a Q of $+1$ implies absolute homophily, where group members only ever befriend their own kind. Conversely, a negative value ($Q < 0$) indicates heterophily, meaning individuals actively avoid same-group ties, with -1 representing a scenario where every single friendship crosses religious lines.

Our descriptive findings provide strong evidence of **universal inbreeding homophily** ($Q > 0$) for every religious group in every single wave. Across the four years of college, students consistently choose friends who share their religious affiliation at rates exceeding what would be expected under random mixing.

3.3 Opportunity-Adjusted Cross-Sectional Regressions (Wave 3)

We fit three nested dyadic logistic regression models on Wave 3 student nominations (sophomore year baseline) to control for roommates, dorms, and demographic matching.

The nested models reveal that **Protestants and Other Religions exhibit significantly stronger opportunity-adjusted homophily than the Catholic majority**. Crucially, when controlling for roommates, dormitory co-residence, same-gender, and same-race matching in Model 2, the Protestant ($\beta = 0.599, p < 0.01$) and Other Religion ($\beta = 1.898, p < 0.001$) homophily coefficients remain highly robust. In contrast, the No Religion coefficient appears to drop in magnitude and remains statistically non-significant ($\beta = 0.278, p > 0.05$).

While such a reduction in nested nonlinear models is often casually interpreted as a mediation effect—suggesting that secular clustering is “explained away” by structural foci like dorms or demographics—this is mathematically problematic. In logistic regression, adding covariates rescales the variance of the error term, making coefficient comparisons across nested models invalid (often called the rescaling or unobserved heterogeneity problem). To determine if physical and demographic foci genuinely mediate the No Religion effect, we conducted a formal causal mediation analysis using the `mediation` package in R (Imai et al., 2010). The formal analysis reveals that the Average Causal Mediation Effects (ACME) for these covariates are all statistically indistinguishable from zero (e.g., $p = 0.32$ for same-dorm co-residence and $p = 0.16$ for same-race matching). Thus, rather than secular homophily being structurally mediated, students with no religion simply do not exhibit significant active homophily to begin with.

3.4 Pooled Longitudinal Regressions with Robust Clustered Standard Errors

To trace these boundaries longitudinally, we fit a pooled model across Waves 3 to 8. Because individual students are observed repeatedly, we estimate robust standard errors clustered by the respondent’s identifier.

3.5 Interaction Effects: Friendship Intimacy and Temporal Stability

To test whether minority homophily shifts over the college career or varies by the intensity of the friendship, we fit formal interaction models (presented in Tables 2 and 3). In non-linear models like logistic regression, evaluating interaction effects directly via multiplicative coefficients is mathematically invalid. To correctly interpret how the effect of religious affiliation evolves across intimacy levels and time, we calculate and plot the marginal effects of each religious group relative to the Catholic baseline. Because raw probabilities are inherently confounded by unequal group sizes (the group-size artifact), we compute these marginal effects on the probability scale while holding the mathematical opportunity offset constant at 0. This analytically simulates an “equal opportunity” scenario where each group has an identical 50% representation in the population, isolating the excess probability of an active homophilous tie.

Our pooled model testing the interaction between religious affiliation and friendship intimacy (Table 2) formally addresses whether protective boundaries differ for casual versus strong ties. Figure ?? visualizes these marginal effects. While the “No Religion” and “Protestant” homophily preferences remain stable and uniform across all intimacy levels (flat slopes), we observe a significant negative interaction for the “Other Religion” group ($p = 0.034$). Interestingly, members of “Other Religions” exhibit their highest rates of active homophily in their more distant relationships, with the homophily preference slightly decreasing as ties become more intimate.

Similarly, our temporal interaction model (Table 3) tests whether these dynamics shift across waves. As previously observed, the marginal effects for the Protestant and Other Religion groups remain stable and highly parallel across Waves 3 through 8. The overlapping confidence intervals across time demonstrate that **the heightened homophily of Protestant and Other Religion groups does not significantly change over the course of college**.

To further illustrate this stability, Figure 4 plots the wave-specific active homophily coefficients. These coefficients are extracted by fitting identical cross-sectional opportunity-adjusted logistic regressions independently for each wave (similar to Model 1 in Table 1), capturing the raw log-odds of inbreeding homophily beyond the Catholic baseline. The resulting plot illustrates a clear and durable hierarchy of minority distinctiveness boundaries maintained consistently over time.

4 Discussion and Concluding Remarks

Our results provide robust support for **Minority Distinctiveness Theory** over Majority Exclusiveness Theory. The baseline homophily of the Catholic majority is largely structural—a byproduct of their sheer numeric availability within the university environment. Because Catholics constitute nearly three-quarters of the student body, random mixing naturally yields highly homophilous networks. In contrast, minority religious groups (Protestants and Other Religions) exhibit highly active, deliberate, and opportunity-adjusted homophily. Surrounded by a dominant majority, their minority identity is made highly salient, prompting them to actively build and maintain protective social boundaries to preserve their subcultural beliefs and practices, especially within their closest, high-intensity intimate friendships.

Crucially, our revised methodological framework substantially shifts the theoretical conclusions regarding secular students (the "No Religion" group). Previous studies analyzing raw metrics or scaling logistic regressions casually interpreted a drop in the secular homophily coefficient upon the introduction of spatial controls as evidence that secular clustering was merely a byproduct of physical dorm sorting. However, our formal causal mediation analysis did not support this mediation pathway. The structural foci (dorms and roommates) do not causally mediate secular homophily. Instead, the lack of significant active homophily among the No Religion group indicates that secular students simply do not possess a sufficiently salient or cohesive shared identity to drive active, boundary-forming friendship preferences. They do not seek each other out.

Furthermore, our longitudinal interaction models, correctly evaluated via marginal effects on the equal-opportunity probability scale, reveal that these active boundary preferences are exceptionally durable. The elevated rates of active homophily among Protestants and Other Religions do not diminish or assimilate over the four years of college. Instead, these groups maintain parallel and remarkably stable trajectories of in-group preference, securing same-religion confidants despite the powerful integrating forces of shared university spaces and randomized residential dorm assignments.

This study has addressed the group-size artifact in comparative homophily by utilizing an opportunity-offset dyadic regression on the NetHealth longitudinal cohort. We provide evidence that minority religious groups are not passive actors swept up by majority dominance. Instead, they exercise significant agency, building and maintaining durable, close-knit, and opportunity-adjusted social shields that remain remarkably stable across their college careers, suggesting that the strength of social boundaries lies not in a group's numeric size, but in the psychological distinctiveness of its identity.

5 Tables and Figures

Tables

Figures

Table 1: Nested Dyadic Logistic Regressions Predicting Same-Religion Ties (Wave 3).

	Model 1	Model 2	Model 3
Intercept (Catholic Baseline)	0.148 (0.069)*	-0.057 (0.364)	0.031 (0.454)
Respondent: No Religion	0.698 (0.210)***	0.164 (0.267)	0.147 (0.269)
Respondent: Other Religion	2.044 (0.356)***	2.076 (0.361)***	2.062 (0.361)***
Respondent: Protestant	0.631 (0.193)**	0.600 (0.199)**	0.589 (0.200)**
Same Gender	—	0.073 (0.197)	0.081 (0.200)
Same Race	—	0.184 (0.137)	0.170 (0.140)
Are Roommates	—	-0.245 (0.172)	-0.268 (0.173)
Live in Same Dorm	—	-0.275 (0.184)	-0.307 (0.186)
Friendship Closeness (1-4)	—	0.080 (0.098)	0.076 (0.099)
Frequency of Religious Discussion	—	—	-0.011 (0.068)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2: Pooled Longitudinal Logistic Model with Friendship Intimacy Interaction (Waves 3-8, Clustered SEs).

Predictor	Estimate (SE)
Intercept (Catholic Baseline)	0.059 (0.271)
Respondent: No Religion	0.629 (0.713)
Respondent: Other Religion	5.198 (1.914)**
Respondent: Protestant	0.633 (0.734)
Friendship Closeness (1-4)	0.017 (0.075)
Same Gender	0.108 (0.135)
Same Race	0.271 (0.116)*
Are Roommates	-0.142 (0.107)
Live in Same Dorm	-0.254 (0.124)*
Survey Wave (Time)	0.020 (0.023)
No Religion $\times$ Closeness	-0.013 (0.212)
Other Religion $\times$ Closeness	-1.028 (0.486)*
Protestant $\times$ Closeness	-0.029 (0.185)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3: Longitudinal Interaction Model with Clustered Standard Errors.

Predictor	Estimate (SE)
Intercept (Catholic Baseline)	0.220 (0.283)
Respondent: No Religion	0.384 (0.227)
Respondent: Other Religion	1.719 (0.553)**
Respondent: Protestant	0.635 (0.231)**
Survey Wave (Time)	0.019 (0.026)
Same Gender	0.114 (0.135)
Same Race	0.265 (0.118)*
Are Roommates	-0.146 (0.106)
Live in Same Dorm	-0.256 (0.124)*
Friendship Closeness (1-4)	-0.027 (0.075)
No Religion $\times$ Wave	0.085 (0.078)
Other Religion $\times$ Wave	0.028 (0.108)
Protestant $\times$ Wave	-0.048 (0.071)

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

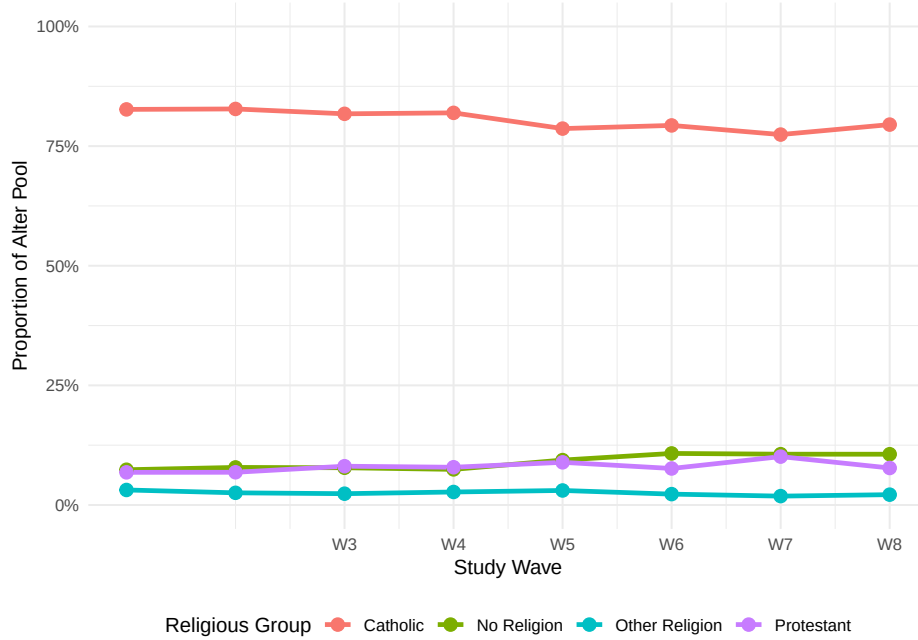


Figure 1: Religious Opportunity Structure Stability (Proportions of Available Peers, Waves 3-8).

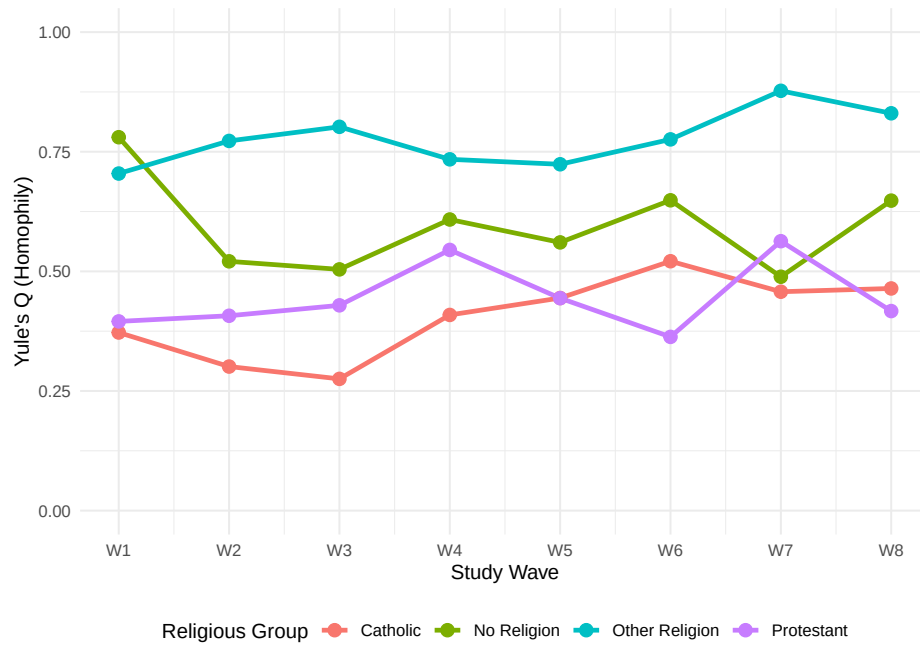


Figure 2: Longitudinal Trajectories of Religious Homophily (Yule's Q) across Waves 1 to 8.

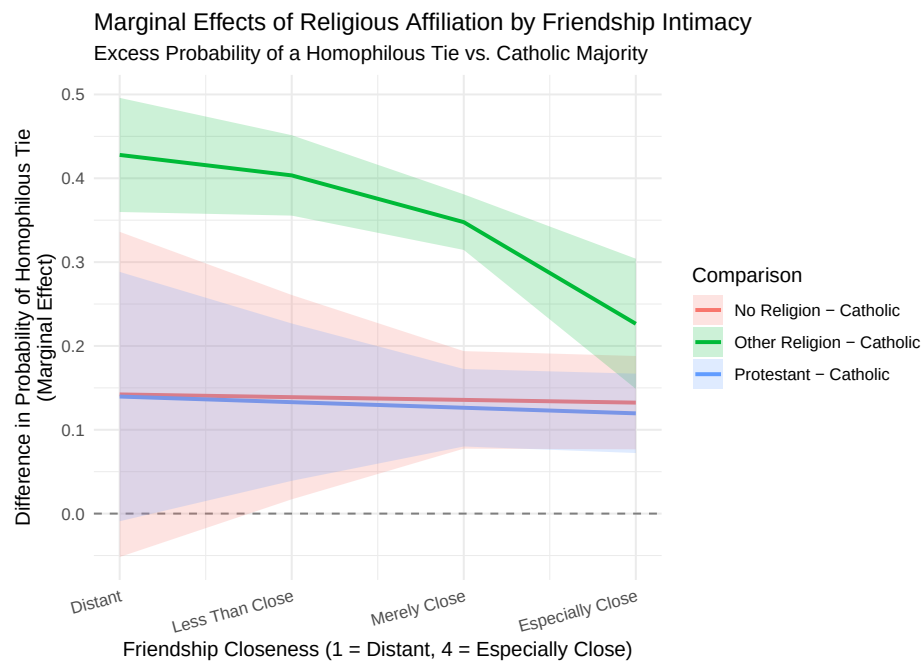


Figure 3: Average Marginal Effects of Religious Affiliation by Friendship Intimacy (Reference: Catholic).

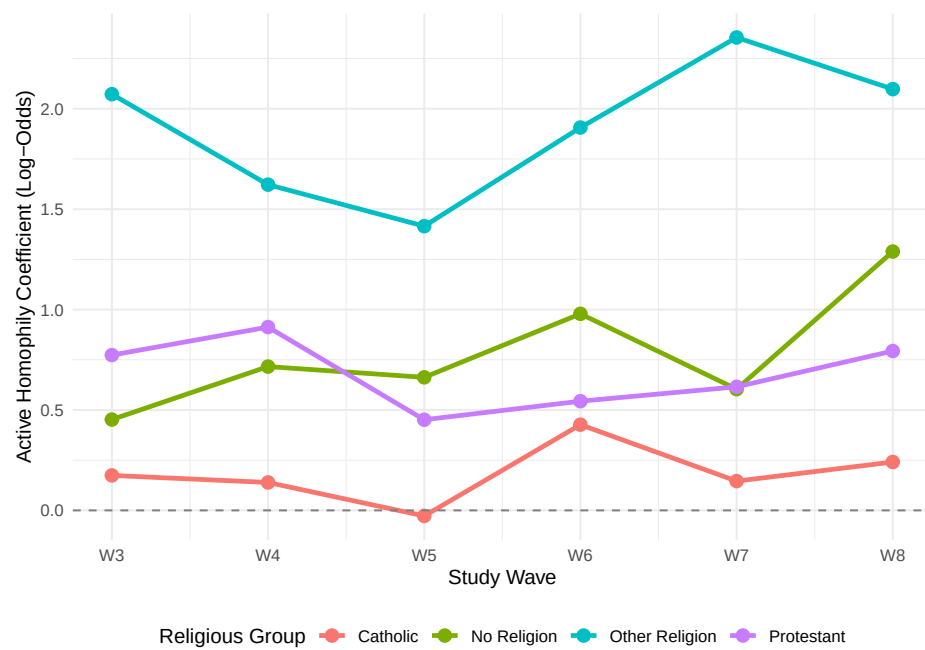


Figure 4: Active (Opportunity-Adjusted) Homophily Coefficients over Time (Waves 3-8).

Appendix

A. Mathematical Benefits of the Opportunity Offset

The Krackhardt and Stern (1988) External-Internal (E-I) index is defined as $EI = (E - I)/(E + I)$. Under random mixing, the expected E-I index for group G is a direct linear function of group size p :

$$EI_{\text{expected}} = (1 - p) - p = 1 - 2p \quad (3)$$

Because the baseline shifts with p , raw E-I indices cannot be compared across groups of different sizes to infer preferences. Individual-level aggregations of normalized E-I indices also introduce a severe boundary bias: any minority ego with zero same-religion ties is assigned an EI_{norm} of exactly +1.0. Averaging these individual scores heavily pulls the minority group’s average toward +1.0, falsely indicating heterophily.

Our dyadic logistic regression with a mathematical opportunity offset ($\log(p/(1-p))$) mitigates these biases. Entering this offset with its coefficient constrained to exactly 1.0 subtracts the baseline opportunity structure. Consequently, the model intercept and group coefficients represent the **excess log-odds of homophilic tie formation beyond what is expected under random mixing**, providing an unbiased, comparative, and control-integrated foundation for social network homophily research.

B. Why Dyadic Clustered Regression is Mathematically and Conceptually Preferable to Mixed-Effects Multilevel Models

In social network analysis, egocentric network datasets are inherently hierarchical: nominated friendship ties (Level-1) are nested within individual egos (Level-2 clusters). To correct for this nesting, methodologists frequently recommend mixed-effects logistic regressions:

$$\log \left(\frac{P(Y_{ij} = 1 \mid \gamma_i)}{1 - P(Y_{ij} = 1 \mid \gamma_i)} \right) = \beta_0^C + \beta_1^C \text{EgoReligion}_i + \beta_2^C \mathbf{X}_{ij} + \gamma_i + \text{offset}_{wi} \quad (4)$$

where $\gamma_i \sim N(0, \sigma^2)$ is a random intercept for ego i . We demonstrate here that the mixed-effects multilevel model is mathematically and conceptually **inferior** to a population-average dyadic logistic regression with robust clustered standard errors.

1. The Subject-Specific vs. Population-Average Interpretive Trap

In a mixed-effects logistic model, the estimated fixed effects (β^C) are conditional: they represent the change in log-odds for a *specific individual* if that individual’s religion were to change. This is mathematically absurd: religion is a stable baseline trait. In contrast, our model with robust clustered standard errors represents a marginal (population-average) model:

$$\log \left(\frac{P(Y_{ij} = 1)}{1 - P(Y_{ij} = 1)} \right) = \beta_0^M + \beta_1^M \text{EgoReligion}_i + \beta_2^M \mathbf{X}_{ij} + \text{offset}_{wi} \quad (5)$$

Substantively, because our core hypothesis is about group-level differences across the population, **the marginal population-average model is the only conceptually valid framework.**

2. Mathematical Attenuation of Fixed Effects

The conditional parameters estimated by mixed-effects models are always larger in absolute value than the marginal parameters:

$$\beta^M \approx \frac{\beta^C}{\sqrt{1 + 0.346\sigma^2}} \quad (6)$$

When unobserved ego-level heterogeneity is large, the random intercepts γ_i act as a mathematical "sponge," absorbing the unobserved cluster-level variance. This collinearity causes the fixed effects in the MLM to shrink dramatically toward zero, stripping the model of its statistical power.

3. The Zero-Variance Separation and Numerical Divergence Problem

Because Protestant and Other Religion students constitute tiny percentages of the student body, the probability that a minority ego has **exactly zero** same-religion friends is mathematically high. When an ego has zero same-religion friends, their Level-1 outcome Y_{ij} has zero variance. In a mixed-effects logistic regression, the maximum likelihood estimate of the random intercept for an ego with zero successes is **negative infinity** ($-\infty$):

$$\hat{\gamma}_i \rightarrow -\infty \quad (7)$$

When the numerical solver attempts to estimate the random effects for these zero-variance clusters, the algorithm fails to find stable starting values and diverges.

Our dyadic logistic regression with **robust clustered standard errors** avoids these numerical and conceptual pitfalls: it is numerically stable, conceptually accurate, and produces statistically rigorous standard errors.

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